

Single pion production in NEUT

Clarence Wret
Including work from
Hayato-san, C. Wilkinson,
D. Roy Varier, S. Abe, ...

NuWro/Wrocław workshop
17 Sep 2026



THE
ROYAL
SOCIETY

IMPERIAL

Department of Physics

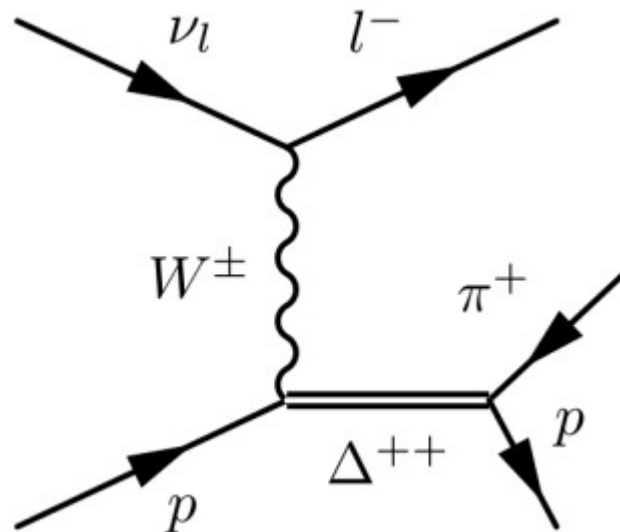
Introduction

- NEUT is primarily the generator of SK and T2K (soon HK)
 - Long lineage on Japanese experiments



(creation date and author) 1988.09.08 ; T.KAJITA CROSS SECTION AT W=1.3-1.4 WITH N(PAI)=2
1993.Dec. ; for Super-Kamioka by Y.Haya... IS ADDED.
(Based on Rein and Sehgal's code)

- Starting seeing use in other experiments for comparisons
- Single pion production, split into two bits: resonant and non-resonant production



$$\nu_l + p \rightarrow l^- + \pi^+ + p$$

$$\nu_l + n \rightarrow l^- + \pi^+ + n$$

$$\nu_l + n \rightarrow l^- + \pi^0 + p$$

$$\nu_l + p \rightarrow \nu_l + \pi^+ + n$$

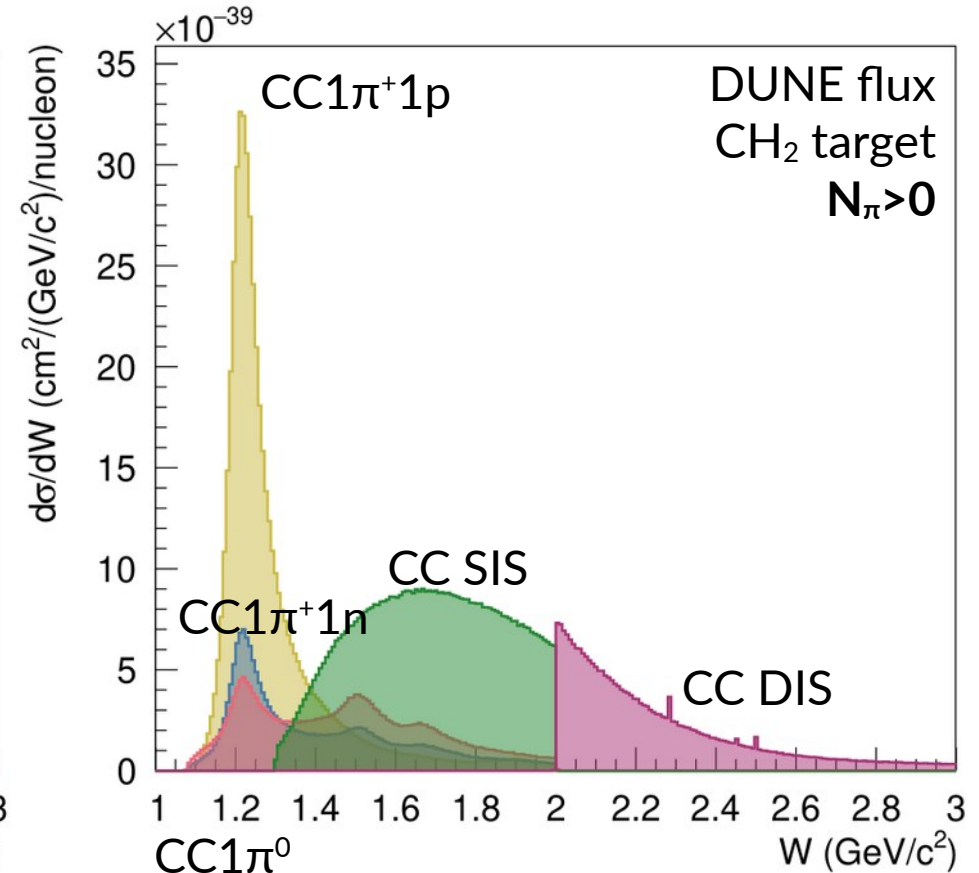
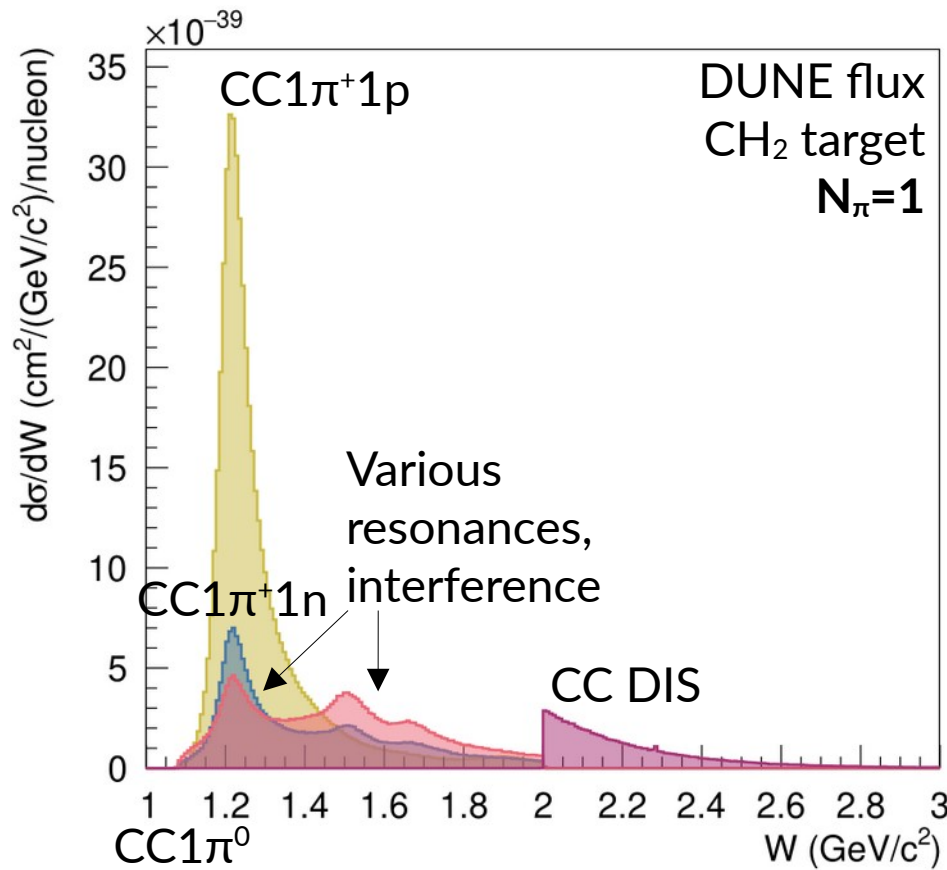
$$\nu_l + p \rightarrow \nu_l + \pi^0 + p$$

$$\nu_l + n \rightarrow \nu_l + \pi^0 + n$$

$$\nu_l + n \rightarrow \nu_l + \pi^- + p$$

Introduction

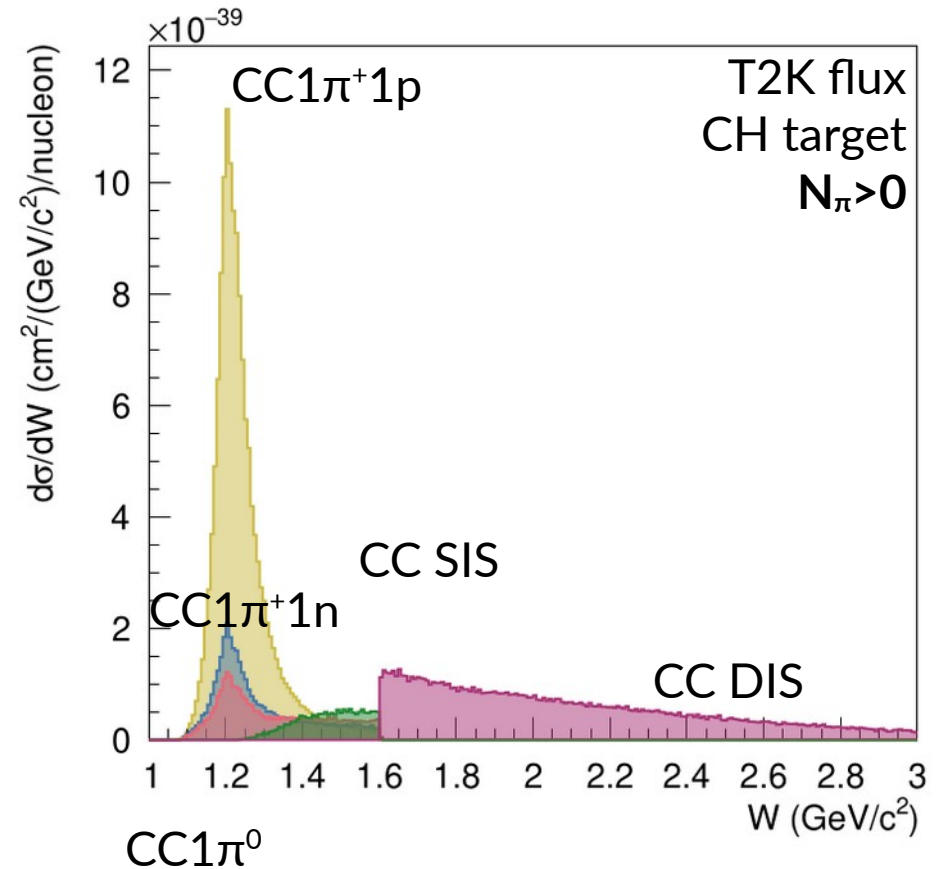
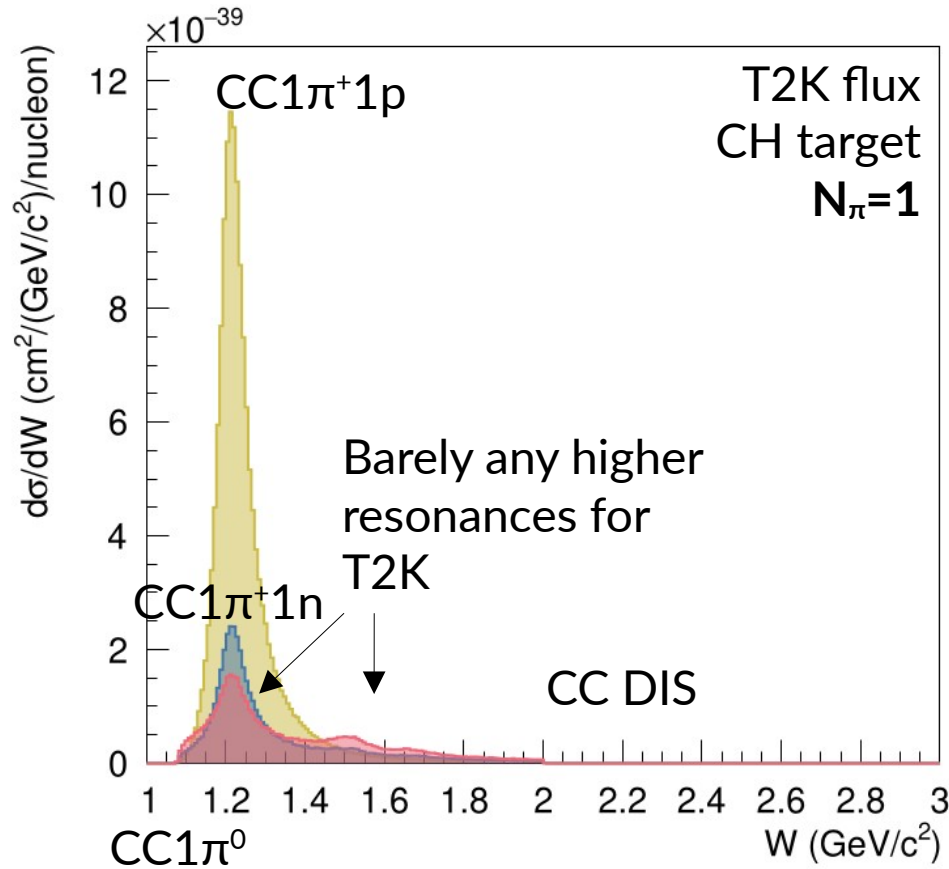
- NEUT uses Rein-Sehgal model (17 resonances; $W < 2.0 \text{ GeV}/c^2$) with numerous modifications



- Non-resonant background from Rein-Sehgal (nucleon)
- Resonance-resonance interference, non-interfering non-resonant background present in mixed isospin channels

Introduction

- NEUT uses Rein-Sehgal model (17 resonances; $W < 2.0 \text{ GeV}/c^2$) with numerous modifications



- Higher resonances matter very little for T2K, but matter more for atmospheric

- Details of Rein-Sehgal implementation
 - PRD 77, 053001 (2008) PRD 90, 093001 (2014)
 - Form factors: Graczyk-Sobczyk (Delta dominated)
 - Lepton mass effects:
(Berger-Sehgal, Graczyk-Sobczyk, Kuzmin et al)
 - PRD 76 113004 (2007) PRD 77, 053003 (2008) Phys. Part. Nucl. 35, S133 (2004)
 - Non-resonant background in mixed isospin channels (no $I_{3/2}$)
 - Delta decay $\rightarrow$ Pion+nucleon ejection mechanism modified
- What GENIE calls “Berger-Sehgal” or “Rein-Sehgal” is not what NEUT calls it
 - e.g. form factors are different, lepton mass effects implemented differently, GENIE ignores resonance interference, very different non-resonant background, Delta decay different
- Fun fact: Hayato-san implemented Rein-Sehgal in 1993, using code directly from Rein and Sehgal (and yes, we have found bugs)

Rein-Sehgal comments

- Rein-Sehgal is based on FKR approximation (quark harmonic oscillator)

PRD 3, 11 (1971)

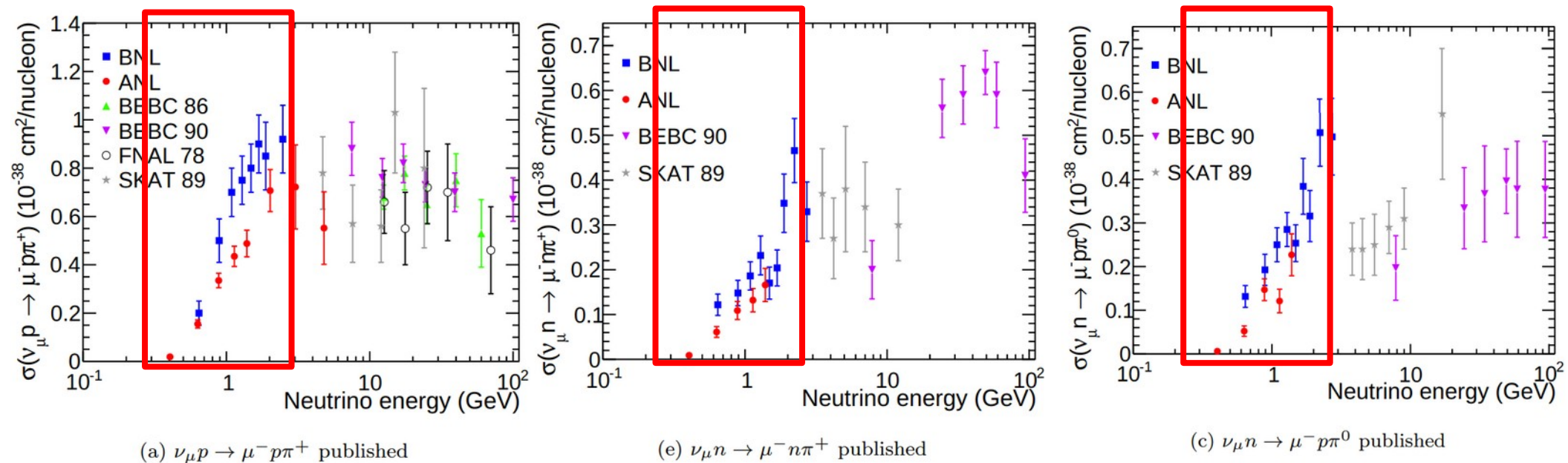
We shall choose a relativistic theory which is naive and obviously wrong in its simplicity, but which is definite and in which we can calculate as many things as possible – not expecting the results to agree exactly with experiment, but to see how closely our “shadow of the truth” equation gives a partial reflection of reality. In our attempt to maintain simplicity, we shall evidently have to violate known principles of a complete relativistic field theory (for example, unitarity). We shall attempt to modify our calculated results in a general way to allow, in a vague way, for these errors.

Starting only from these two constants, the slope of the Regge trajectories, and the masses of the particles, 75 matrix elements are calculated, of which more than $\frac{3}{4}$ agree with the experimental values within 40%.

- Rein-Sehgal non-resonant background?
- Delta domination, do we need higher resonances?
 - Importantly, it doesn't necessarily get these resonances right (signs, FKR formalism with quark harmonic oscillator, etc)
- Arbitrary electron scattering form factors in Rein-Sehgal

Hydrogen data issues

Eur.Phys.J.C 76 (2016) 8, 474



- Differences between ANL and BNL's measurements of CC1 π has been plaguing the field
 - GiBUU, NuWro, NEUT tuning, theory dev, even low- v analyses
 - Corrections exist, but rely on GENIE CCQE model, and no recipe to correct differential cross sections or event rates
 - Hard to say which theory is right when the data isn't reliable
- And data applies **interesting** corrections, and unclear how much data is D₂ and how much is H₂

Fitting philosophy and idea

- Can't use form factor from G&S papers' fits since the base model is different
$$C_5^A(Q^2) = \frac{C_5^A(0)}{(1 + \frac{Q^2}{M_A^2})^2},$$
 - So we run our own fits: M_A , $C_5^A(0)$, non-resonant bkgd
- Intention of fit is **not** to get small uncertainties
 - Consider multiple combinations of data sets and ensure the model and uncertainties **covers reasonable range of data**
 - We'd be very surprised if model fits all the data
→ aim to span all data with appropriate uncertainties

ANL + BNL with $W < 2.0$ GeV with and without $I_{1/2}$ free

ANL only with $W < 2.0$ GeV with and without correction

ANL only with $W < 1.4$ GeV with and without $I_{1/2}$ free with all possible data

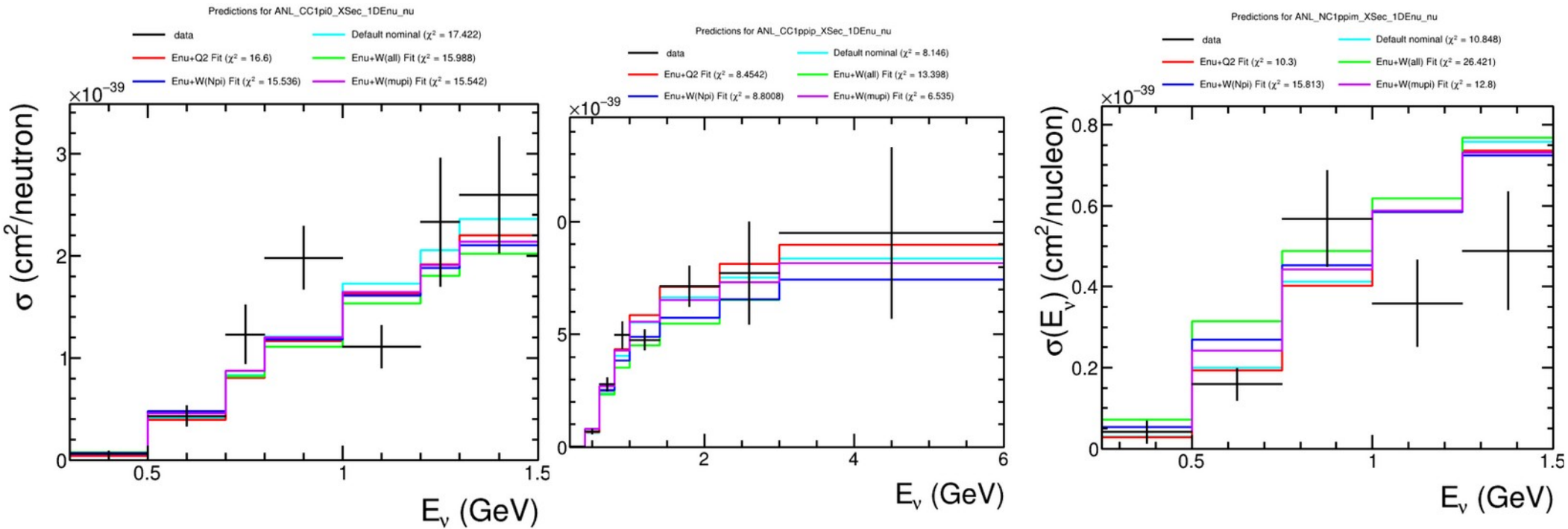
ANL only with $W < 1.4$ GeV without $I_{1/2}$ free, $\sigma(E_\nu)$ and $N(Q^2)$

BNL only with $W < 2.0$ GeV and without correction

MiniBooNE $T_\mu \cos \theta_\mu$ with and without $I_{1/2}$ free, no normalization

Example outputs

- Student (Roy) also tried to include W distributions and pion momentum (shape only) into fit



- This produces many different predictions!

Parameter	Central Value	Uncertainty
$M_A^{RES}(\text{GeV})$	1.07	0.15
C_A^5	0.96	0.15
$I_{1/2}$	0.96	0.40

Parameter in models are not same

- G+S showed that $C_5^A(0)$ correlates with inclusion of D_2 effects; smaller C_5^A without D_2 effects (closer to off-diagonal Goldberger-Treiman relation ~ 1.15)
 - But hard to tell how much data is D_2 and H_2
- Treatment of data also different (ANL+BNL tensions)

	Fit	$C_5^A(0)$	$M_{A\Delta}$ (GeV)	p_{ANL}	χ^2/NDF	NDF
Free n + p	A1	$0.94^{+0.30}_{-0.30}$	$0.93^{+0.18}_{-0.19}$	1.03	0.15	6
	A2	$1.09^{+0.50}_{-0.69}$	$0.94^{+0.30}_{-0.30}$	0.93	1.55	9
	A3	$2.48^{+0.52}_{-0.52}$	$0.75^{+0.14}_{-0.14}$	0.94	1.56	9
	Joint	$0.93^{+0.13}_{-0.13}$	$0.81^{+0.09}_{-0.09}$	0.89	2.11	30
Deuteron	A1	$1.11^{+0.32}_{-0.34}$	$0.97^{+0.17}_{-0.17}$	1.04	0.20	6
	A2	$1.31^{+0.49}_{-0.77}$	$1.00^{+0.27}_{-0.25}$	0.93	1.52	9
	A3	$2.83^{+0.62}_{-0.60}$	$0.76^{+0.13}_{-0.13}$	0.94	1.47	9
	Joint	$1.10^{+0.15}_{-0.14}$	$0.85^{+0.09}_{-0.08}$	0.90	2.06	30

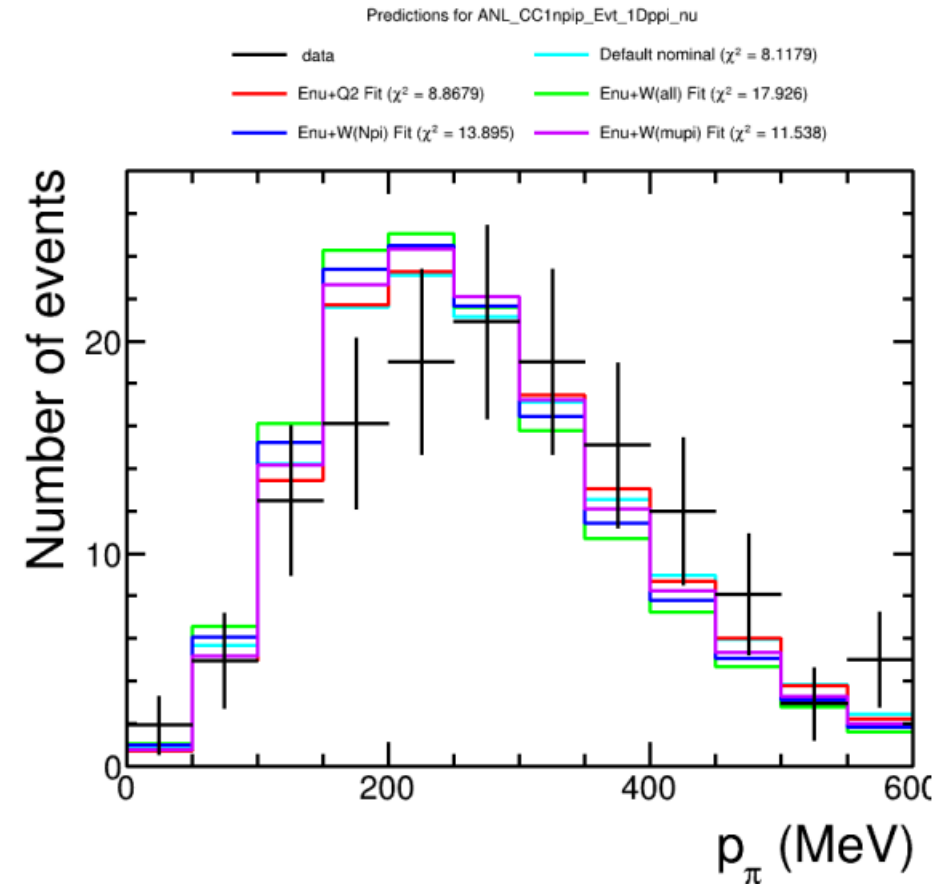
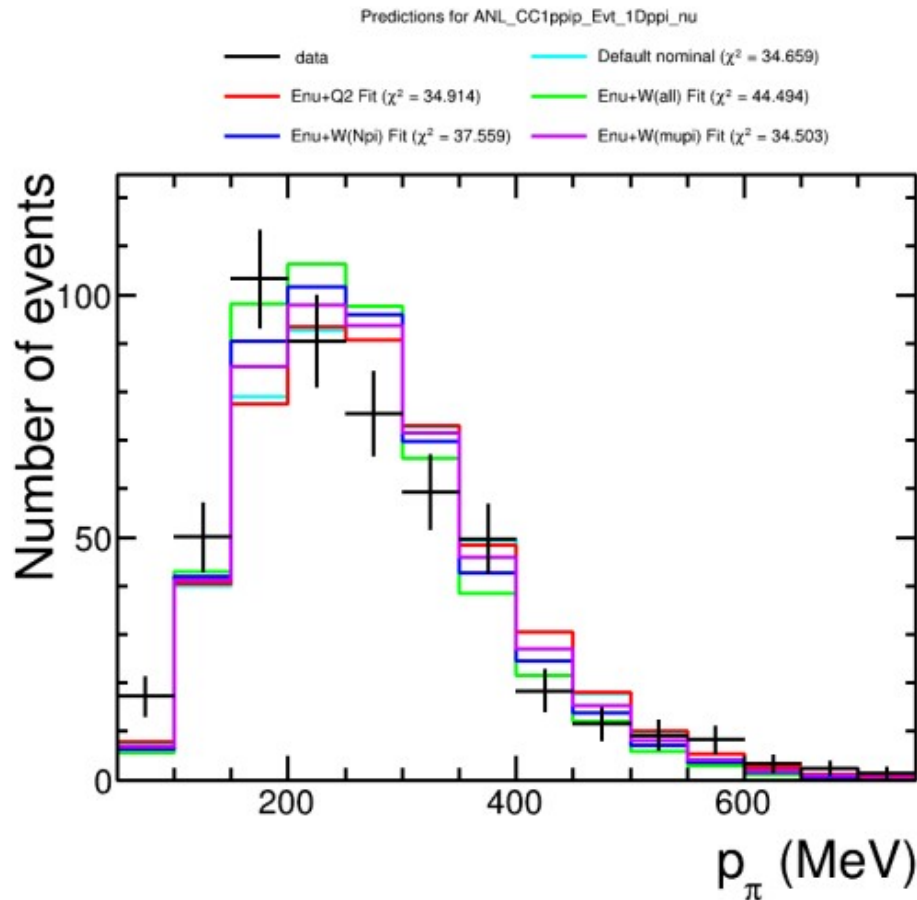
GENIE: M_A^{RES} (GeV/c²) 1.15 ± 0.02 0.88 ± 0.02 1.09 ± 0.014
 T2K: M_A^{RES} (GeV/c²) 1.07 ± 0.15
 GENIE 2: M_A^{RES} (GeV) 0.94 ± 0.05

NuWro
 dipole, only M_A , free target 0.95 ± 0.04

- Note the large difference in uncertainty estimates for NEUT (due to conservative tuning philosophy)

Example outputs, kinematics

- Also studied kinematic distributions



- Fits do impact hadronic observables, but getting large impact on pion momentum means changing xsec a lot
- Did we really consider hadronic freedoms?

Missing freedoms?

- There is freedom in the resonance decay; both if it considers isospin, and uncertainties on the matrix elements from the FKR
- Rein-Sehgal recipe is simple:

$$\frac{d^2\sigma}{dQ^2 dW} \rightarrow \frac{d^2\sigma}{dQ^2 dW} \frac{d\sigma}{d\Omega} = \frac{d^2\sigma}{dQ^2 dW} \frac{d^2\sigma}{d\cos\theta d\phi}$$

- Contract matrix elements with appropriate spherical harmonics
- Isotropic, Delta-only, and multi-resonance
(Delta, $P_{11}(1450)$, $D_{13}(1525)$, $S_{11}(1540)$) decay for element calculation

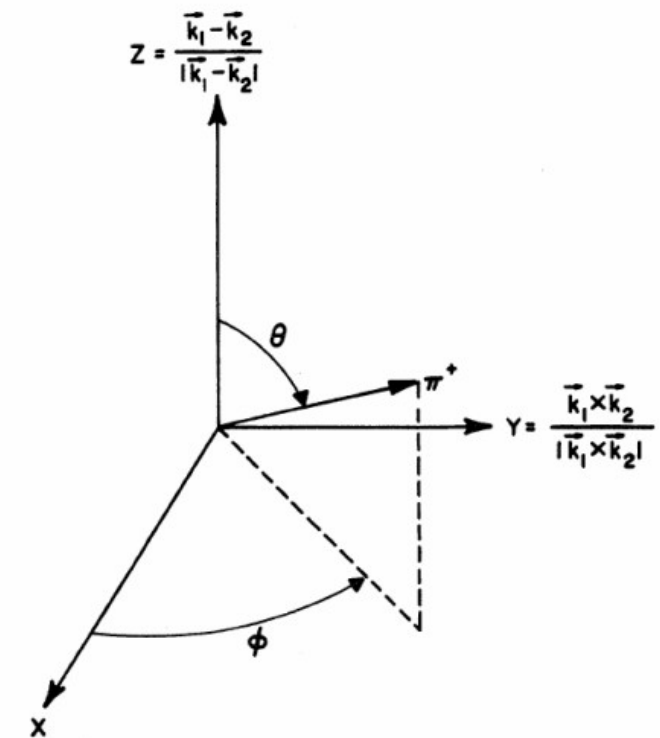


FIG. 14. Definition of the azimuthal ϕ and polar θ angles in the Adler system. $\vec{k}_1$ and $\vec{k}_2$ are vectors along the ν and μ^- directions, respectively, in the $N\pi$ rest system.

Missing freedoms?

- Function to sample the Adler angles θ and ϕ , e.g. for isotropic and Delta-only (multi-res spans pages...)

$$W^{\text{Iso}}(\theta, \phi) = \frac{1}{4\pi} W^{\Delta}(\theta, \phi) = \frac{1}{\sqrt{4\pi}} \frac{1}{\tilde{\rho}} \left\{ Y_0^0 \tilde{\rho} - \frac{2}{\sqrt{5}} Y_0^2 \left(\tilde{\rho}_{33} - \frac{1}{2} \rho \right) + \frac{4}{\sqrt{10}} \left(\text{Re} Y_1^2 \text{Re} \tilde{\rho}_{31} - \text{Re} Y_2^2 \text{Re} \tilde{\rho}_{3,-1} \right) \right\}$$

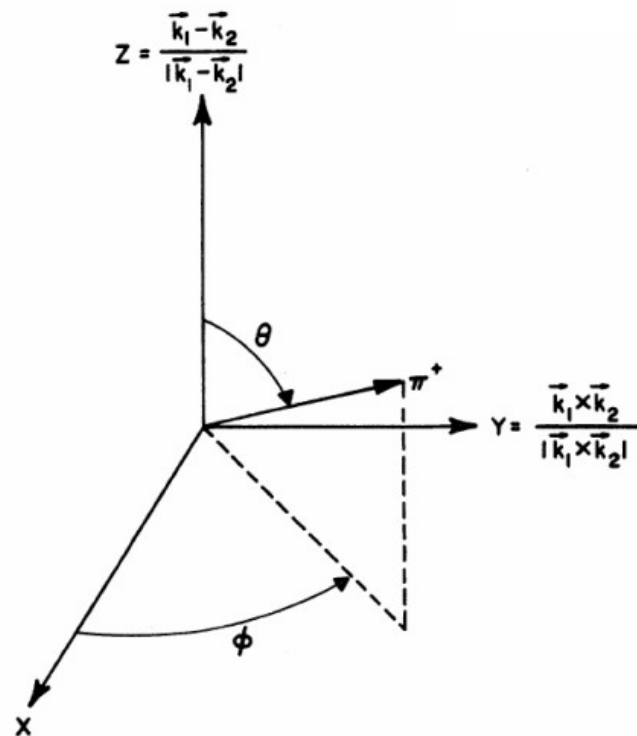
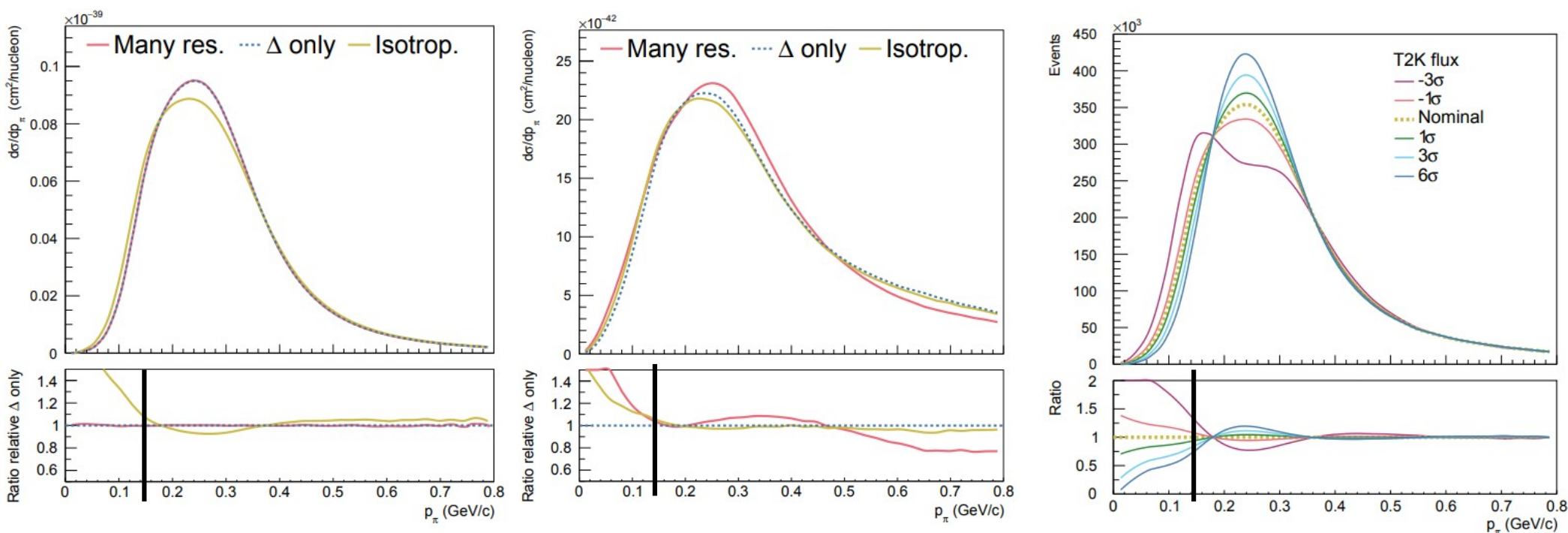


FIG. 14. Definition of the azimuthal ϕ and polar θ angles in the Adler system. $\vec{k}_1$ and $\vec{k}_2$ are vectors along the ν and μ^- directions, respectively, in the $N\pi$ rest system.

- Implemented uncertainties to cover different number of resonances
- And separately also uncertainties on density elements
- Rein-Sehgal and FKR papers

Missing freedoms?

- Checked the impact on T2K flux
 - Variable of interest was pion momentum, since it determines sample size with pions below/above threshold
 - T2K and SK sees excess in data in low pion momentum sample, so was particularly interested in this



Cherenkov threshold for pion

arxiv 2606.14015 (PRD accepted)

Missing freedoms?

- Impacts are also visible on bubble chamber data

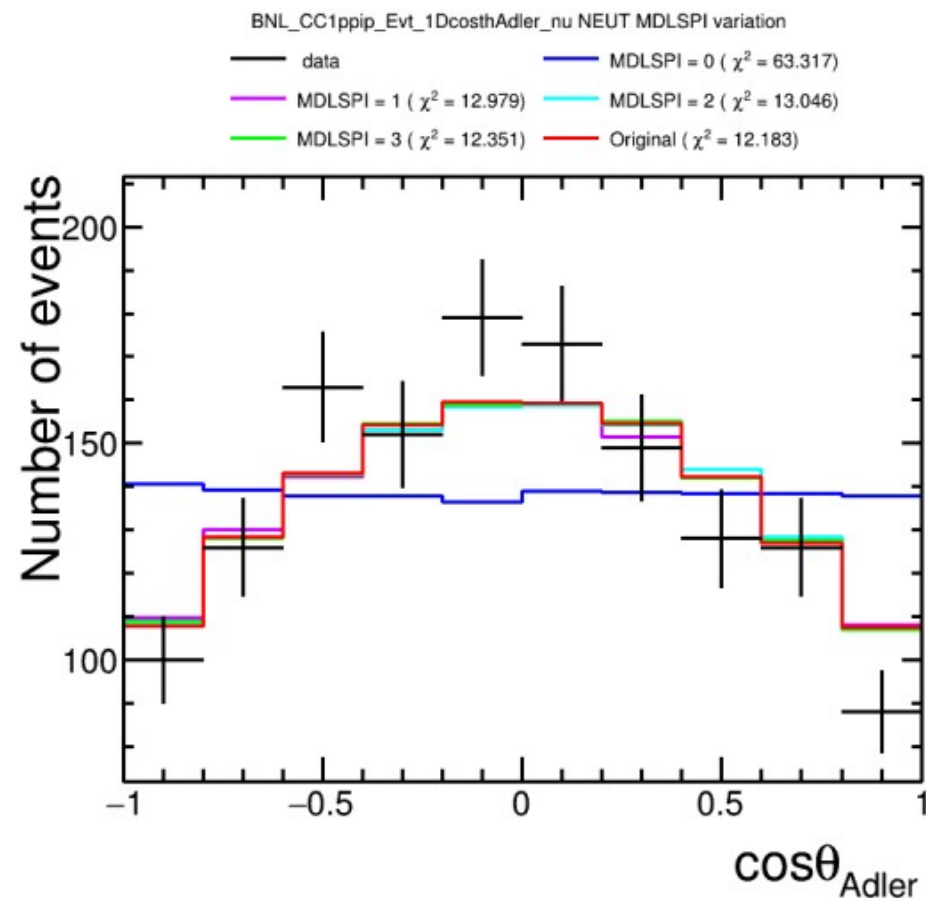
Multi-res clearly different in mixed isospin channel

Isotropic

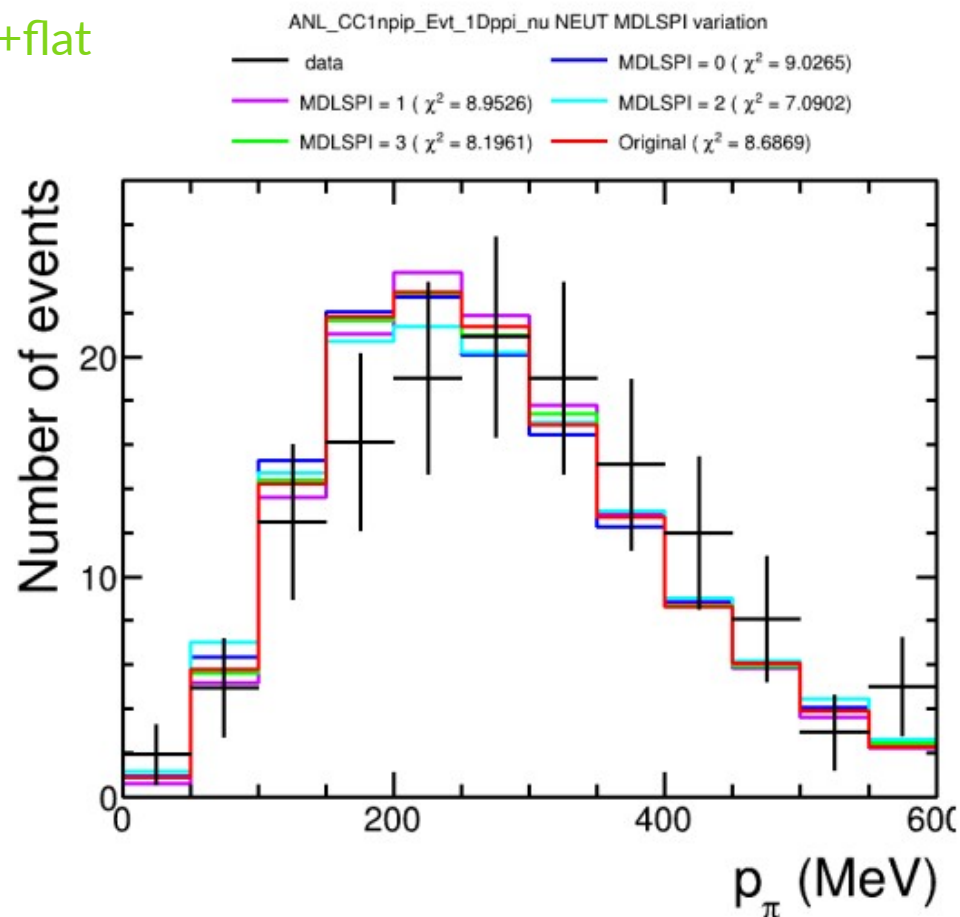
Delta only

Delta+other

Delta+flat



(d) $\nu_\mu d \rightarrow \mu^- p^+ \pi^+$ channel at BNL

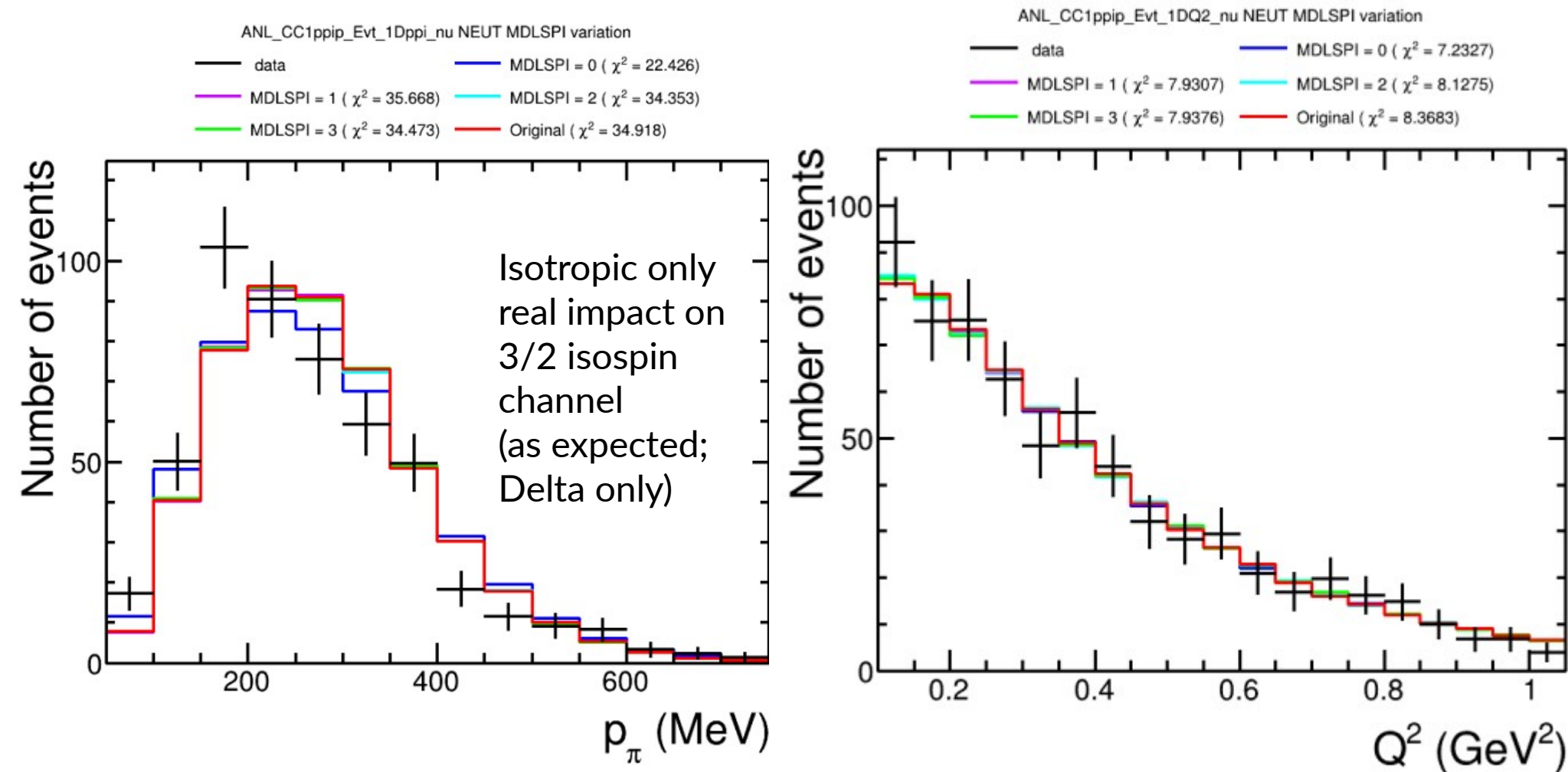


(a) $\nu_\mu d \rightarrow \mu^- n^0 \pi^+$ channel at ANL

- Reminiscent of what Seisho showed for DCC

Missing freedoms?

- As expected, these do not impact Q^2 distribution (by design only impacts resonance decay)



(b) $\nu_\mu d \rightarrow \mu^- p^+ \pi^+$ channel at ANL

New models

- But... you can only get so far with a model based on deeply approximate foundations
- Concerted effort in last few years to include new neutrino-nucleon models in NEUT
 - DCC (see Seisho's talk yesterday)
 - HNV (resonances with $W < 1.4$ GeV)
 - MK (2018, 2025 tuned to ANL+BEBC and BEBC only)
- All significantly improve on Rein-Sehgal model in many different ways
 - e.g. preserve unitarity, not based on FKR, focuses on simpler Delta interaction, considers coupled channels, non-resonant backgrounds...

HNV model

- HNV = **H**ernandez, **N**ieves, **V**alverde; effort from Valencia group to describe single pion interaction in $E_\nu < 2$ GeV
- Follows a long list of papers (and probably earlier)
 - Phys.Rev.D 98 (2018) 7, 073001; Phys.Rev.D 95 (2017) 5, 053007; Phys.Rev.D 93 (2016) 1, 014016; Phys.Rev.D 87 (2013) 11, 113009; Phys.Rev.D 81 (2010) 085046; Phys.Rev.D 76 (2007) 033005
- Original idea was $\Delta(1232)$ with pole contribution, and added background terms from chiral symmetry
- Now also has pole and cross terms, nucleon, contact terms, pion poles, and $D_{13}(1520)$ inclusion
- Useful for the **$W < 1.4$ GeV/c²** range; **relevant to T2K**
- Unitarised by including Watson phases
- This is quite similar to Minoo (but she has many more resonances, so W cut does not apply!)

Why HNV model

- The authors point out that model is generally similar to the DCC model, but with much simpler treatment
- Issue with public DCC model is lack of reweightability (pre-calculated structure functions and amplitudes)
- HNV model is fully exposed to outside elements
 - Can easily change parameter values
 - And calculates the $d\sigma/dQ^2 dW d\cos\theta_{\text{Adler}} d\varphi$
- Probably not the final solution to our pion problems, but a good start!

HNV diagrams

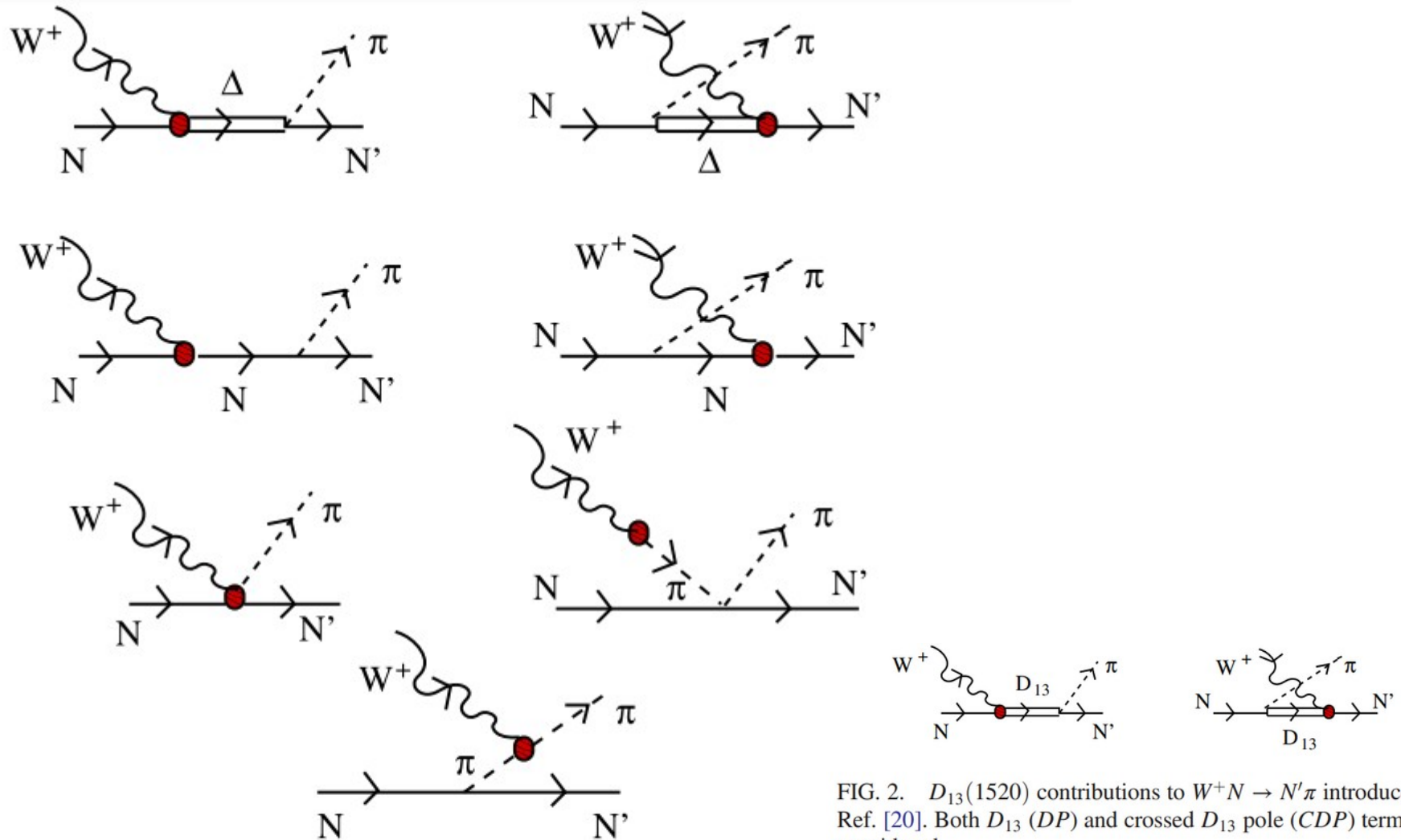
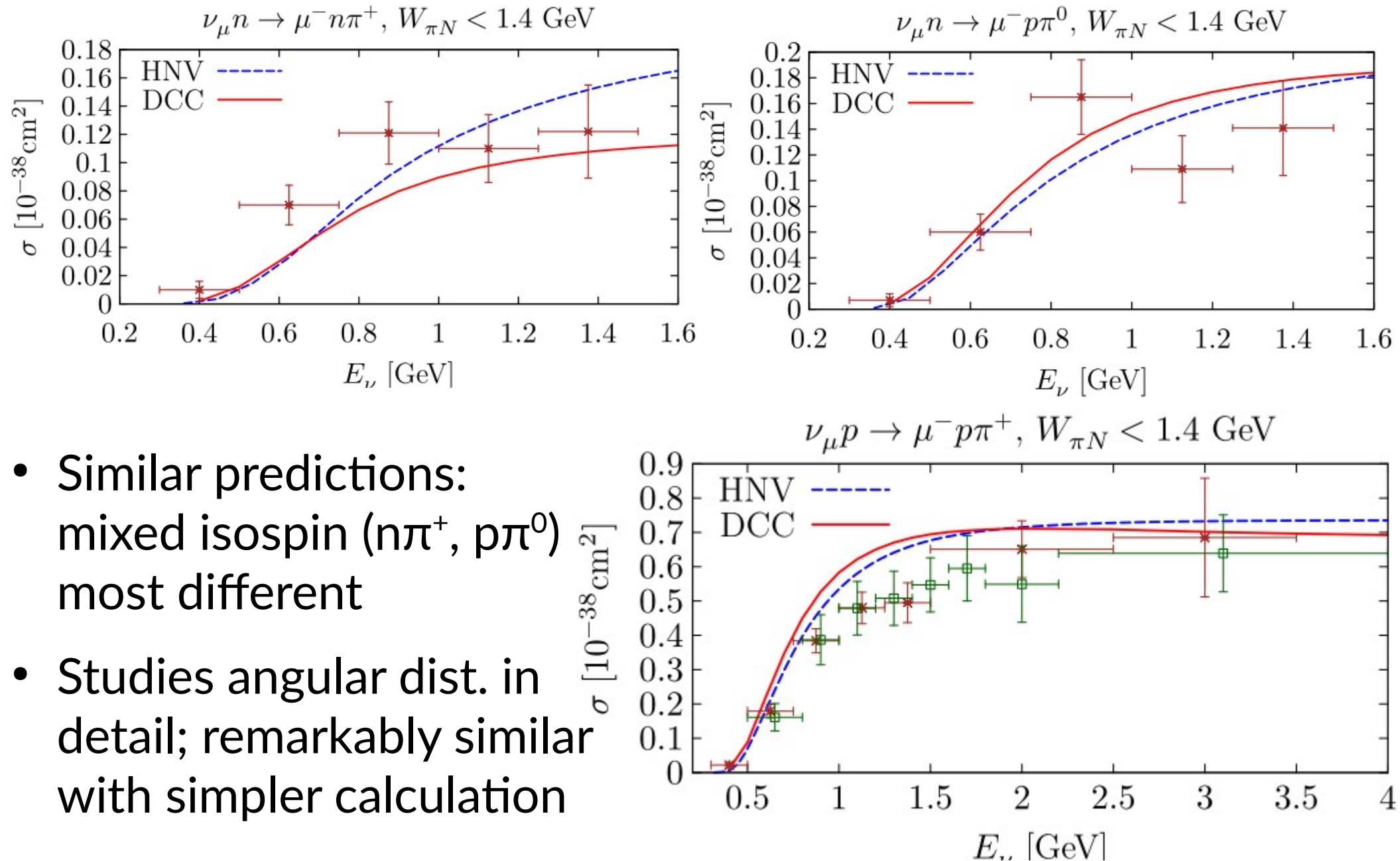


FIG. 2. $D_{13}(1520)$ contributions to $W^+ N \rightarrow N' \pi$ introduced in Ref. [20]. Both D_{13} (DP) and crossed D_{13} pole (CDP) terms are considered.

FIG. 1. Model for the $W^+ N \rightarrow N' \pi$ reaction as introduced in Ref. [10]. It contains the Delta (ΔP) and crossed Delta pole ($C\Delta P$) terms (first row), the nucleon (NP) and crossed nucleon pole (CNP) terms (second row), the contact CT and pion pole (PP) terms (third row), and the pion in flight (PF) term (fourth row).

HNV papers

- Comparisons between DCC and HNV in PRD 98 073001

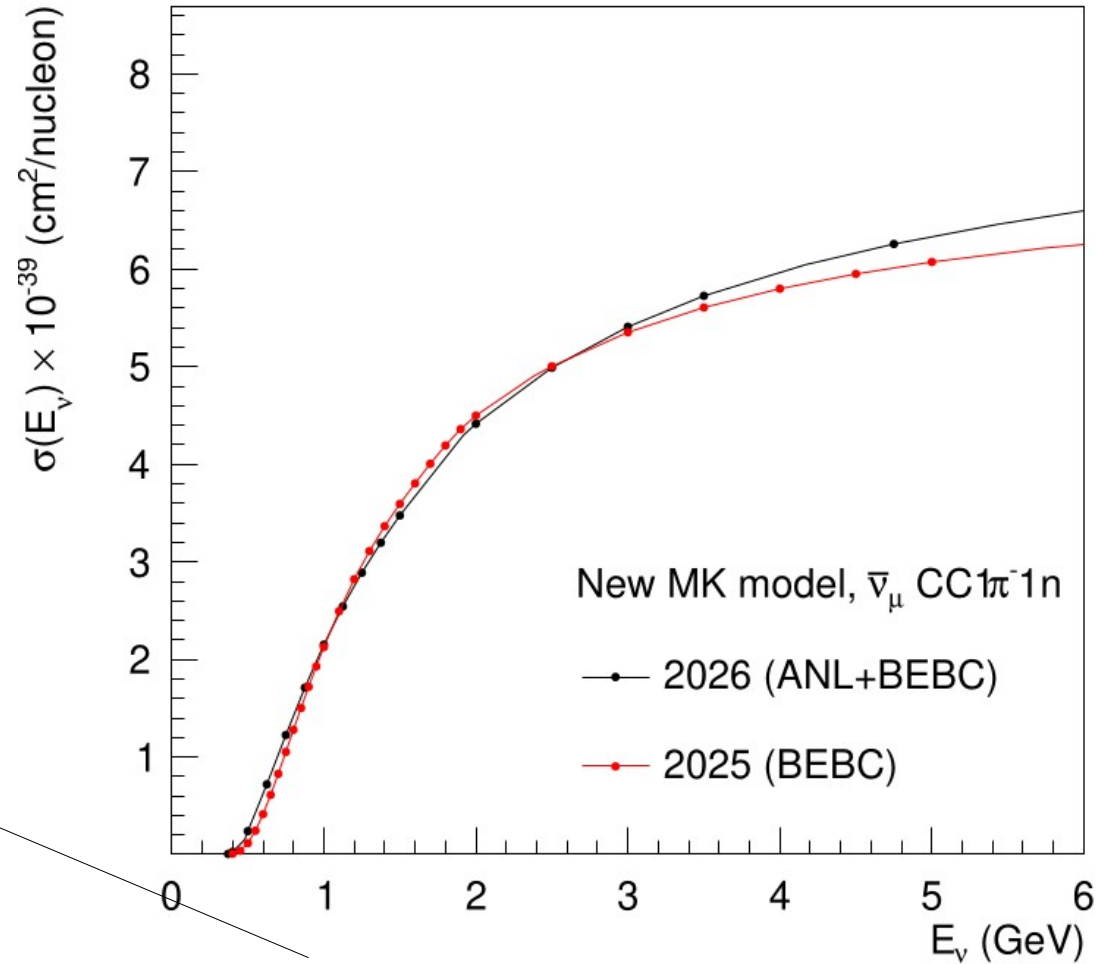
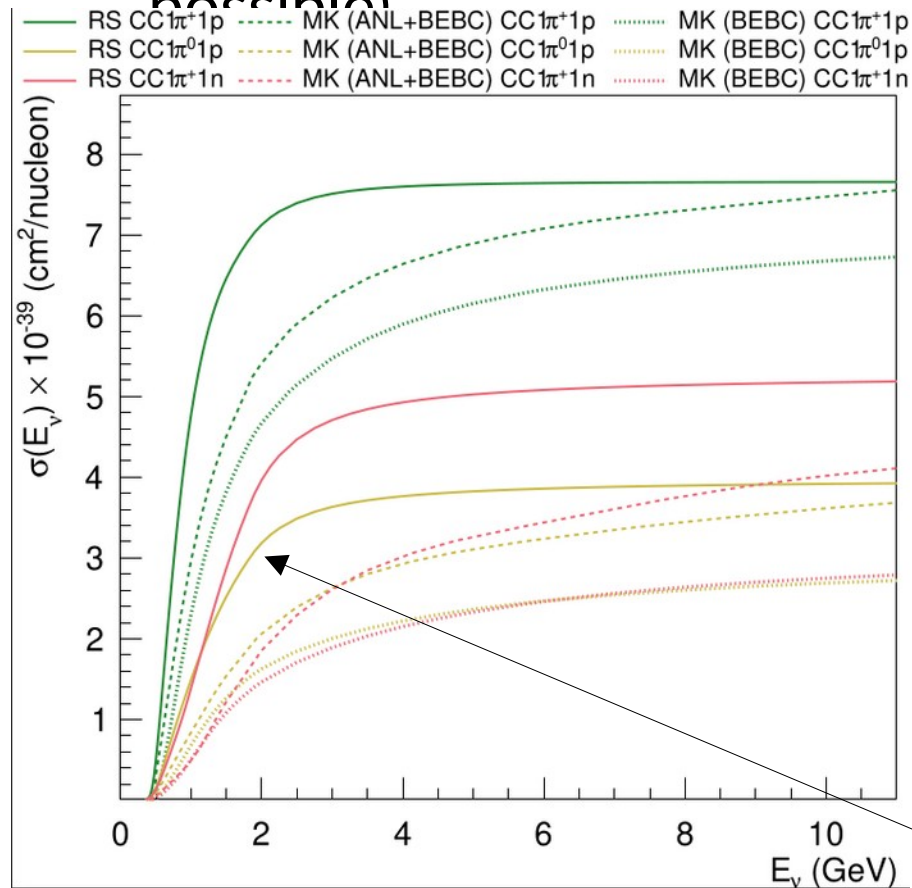


- Large fits to multiple probe datasets (basically does her own MAID analysis, and more)
 - Fitting resonance couplings, phases, vector form factors, etc
 - Fit to over **50,000** non-neutrino data points!

# data point	Photon, electron, pion, Neutrino Channels	Q ² Range (GeV/C) ²	W Range GeV	Form Factors	
≈ 9800	$\gamma p \rightarrow n + \pi^+, \gamma p \rightarrow p + \pi^0$	0	1.08 – 2.0	Proton	Vector
≈ 31000	$ep \rightarrow en + \pi^+, ep \rightarrow ep + \pi^0$	0.16 – 6.0	1.08 – 2.0		
≈ 2500	$\gamma n \rightarrow p + \pi^-$	0	1.08 – 2.0	Neutron	
≈ 700	 $en \rightarrow ep + \pi^-$	0.4 – 1.0	1.08 – 1.8		
≈ 400	$\pi^+ p \rightarrow p + \pi^+, \pi^- p \rightarrow p + \pi^-$	0	1.08 – 2.0	Axial-Vector	
<100	$\nu N \rightarrow l^- N + \pi, \bar{\nu} N \rightarrow l^+ N + \pi$	Q ² >0 Integrated	1.08 – 2.0 Integrated		

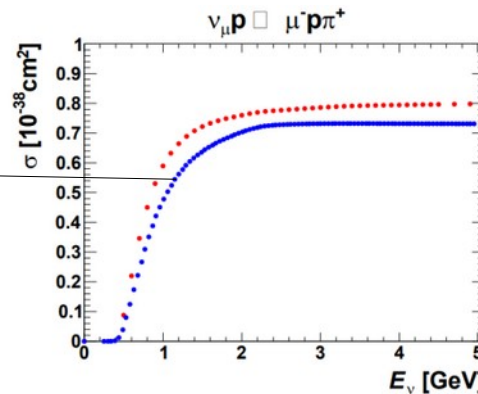
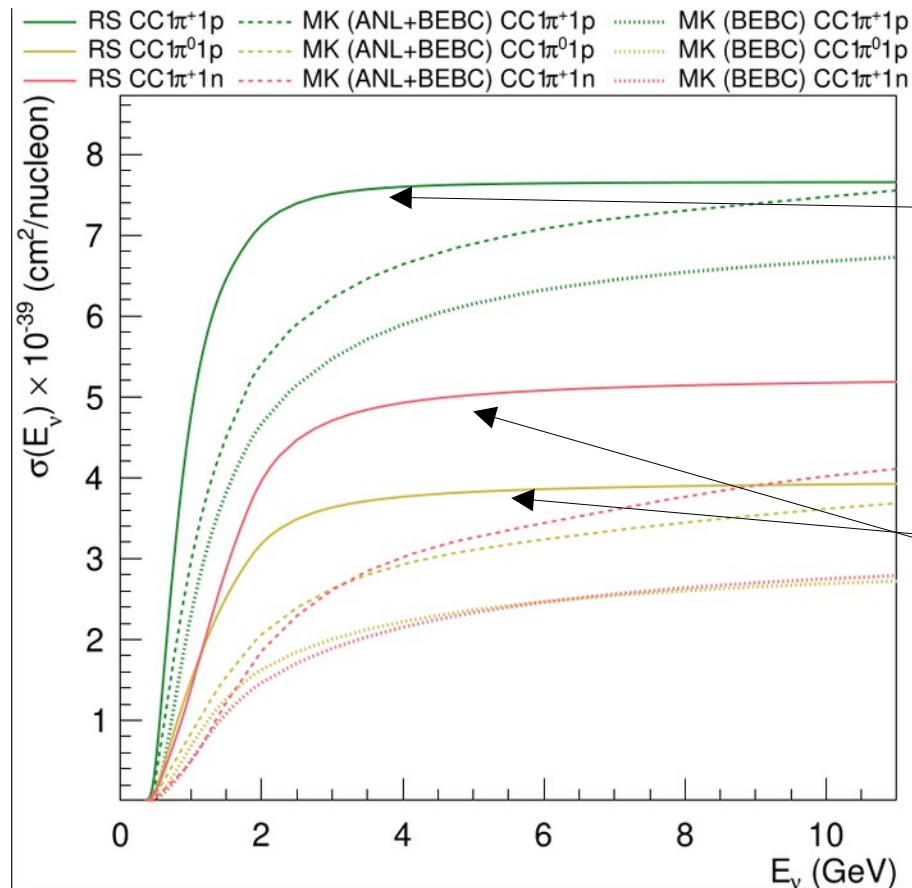
- All resonances present, non-resonant backgrounds, interferences, all resonances generate hadronic system (similar to DCC in this sense)

- Uses ANL and BEBC data for neutrino fit
 - Concerned about deuterium effects in data (wanted pure H₂ if possible)

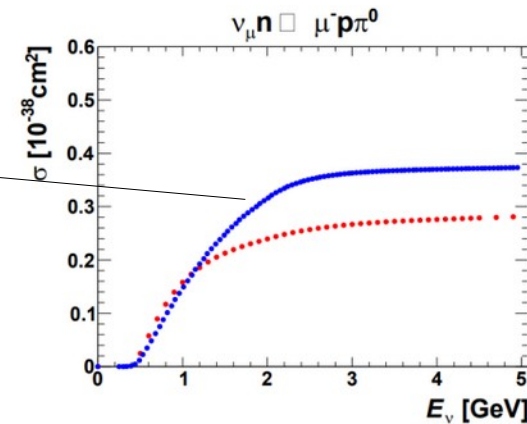
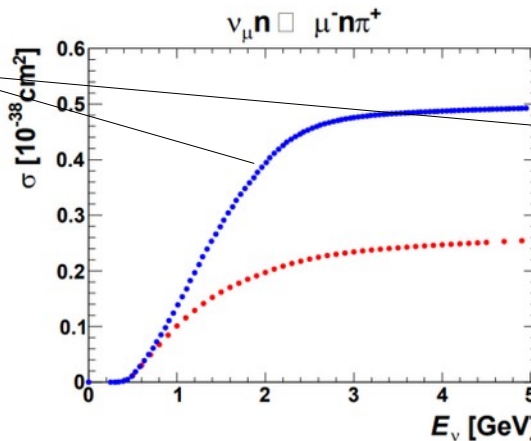


- Cross section much lower than Rein-Sehgal, especially at T2K
 - And anti-neutrino largely decoupled from neutrino

- Cross section much lower than Rein-Sehgal, especially at T2K
 - But DCC is much larger than Rein-Sehgal...



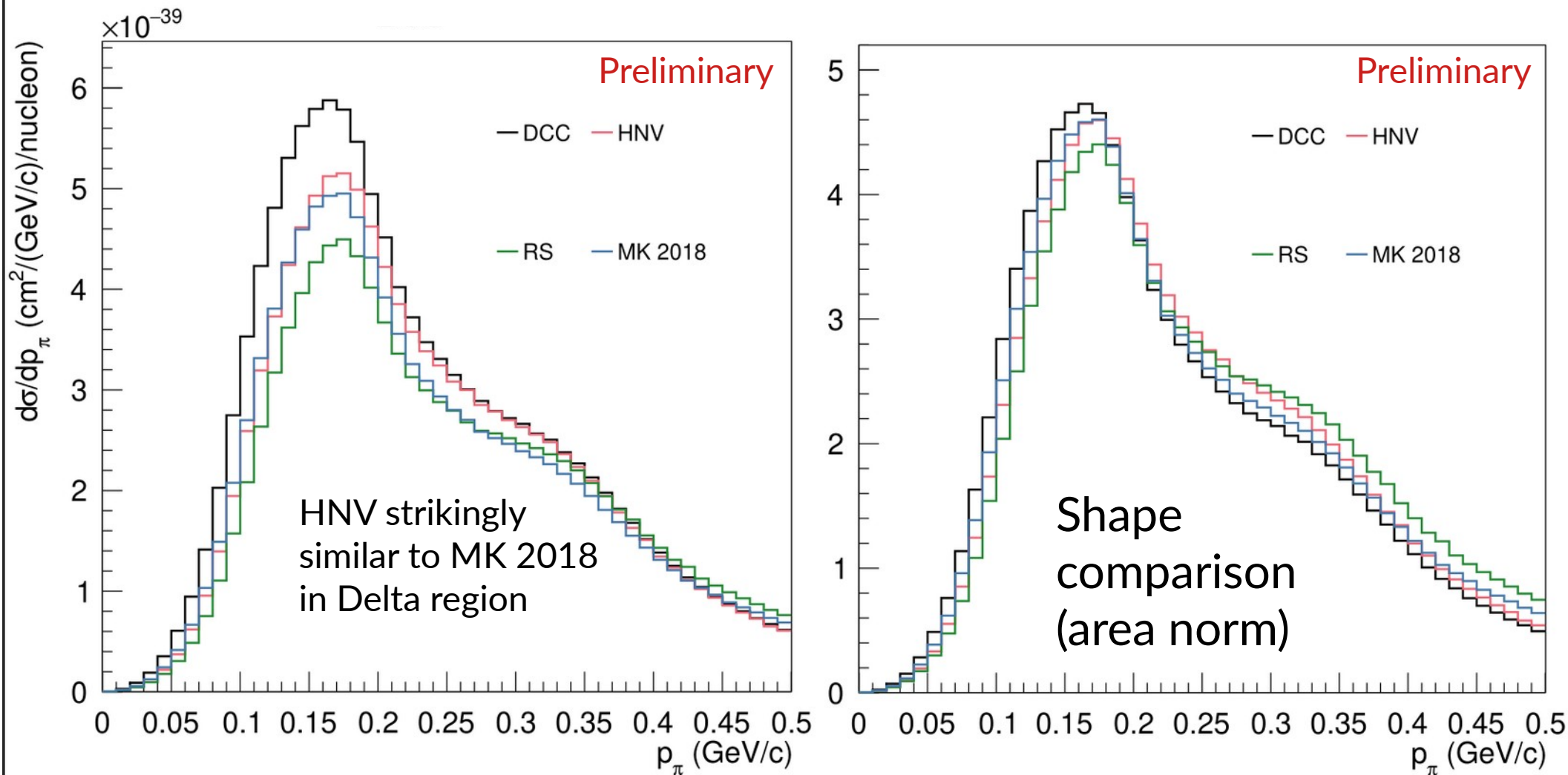
DCC
Berger-Sehgal



- The cross section grows much slower in energy for MK model
 - Pinned by high energy BEBC data, not so much ANL?

Brief comparisons

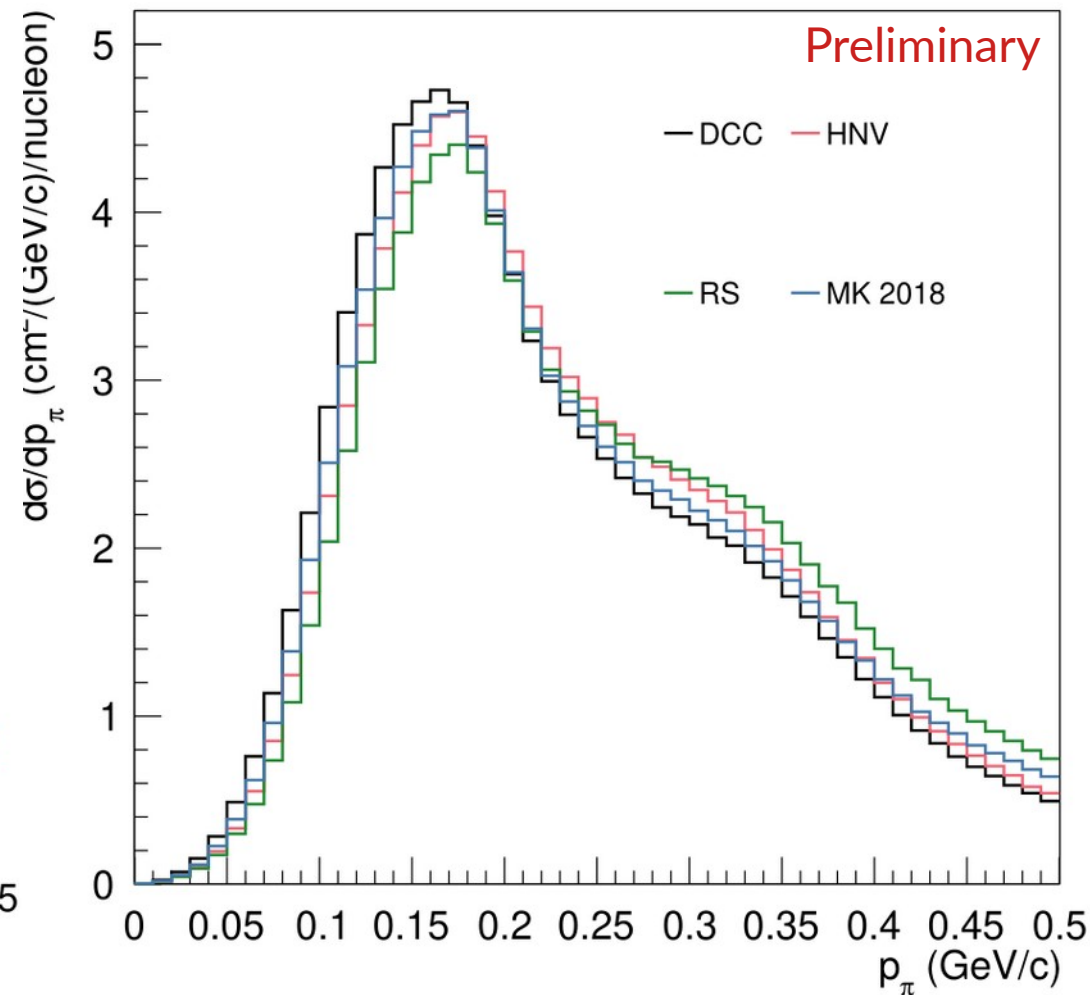
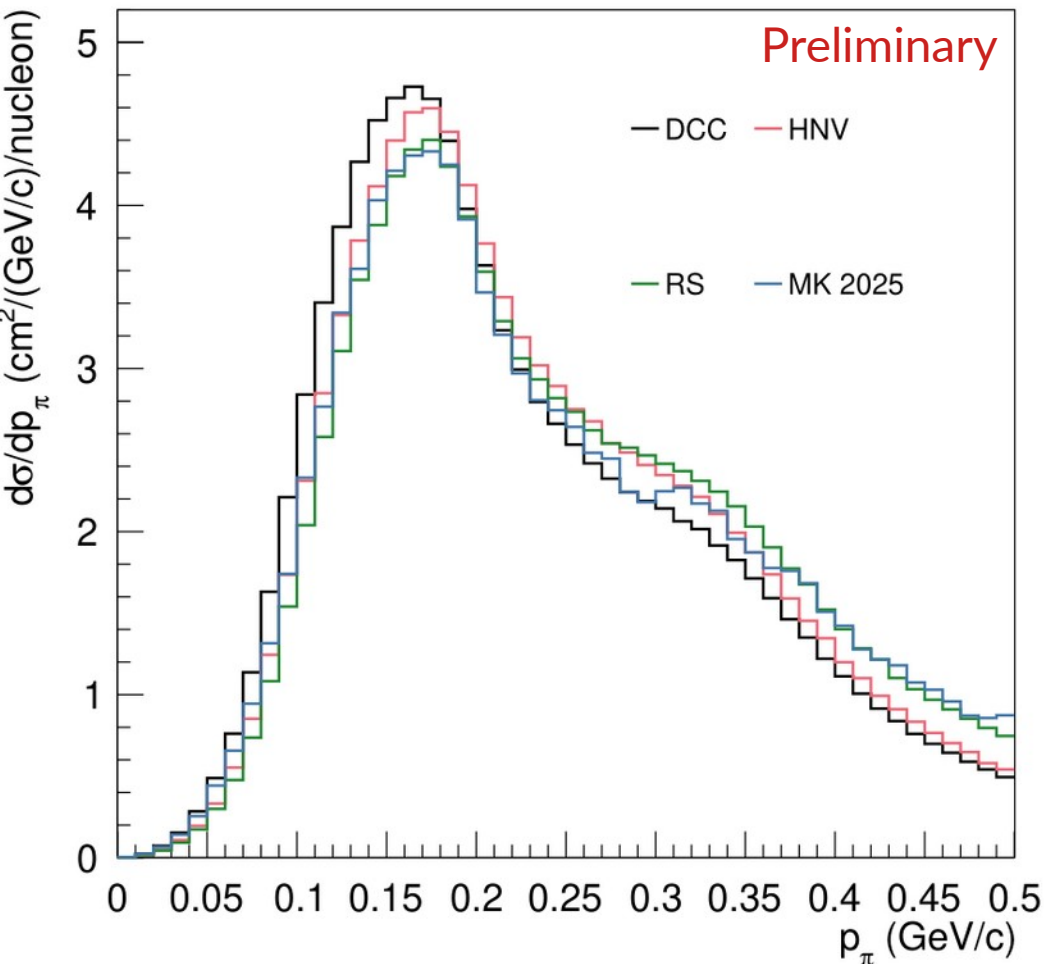
- Am almost done with validating the generators (but not confident enough to show all distributions!)



Brief comparisons

- Am almost done with validating the generators (but not confident enough to show all distributions!)

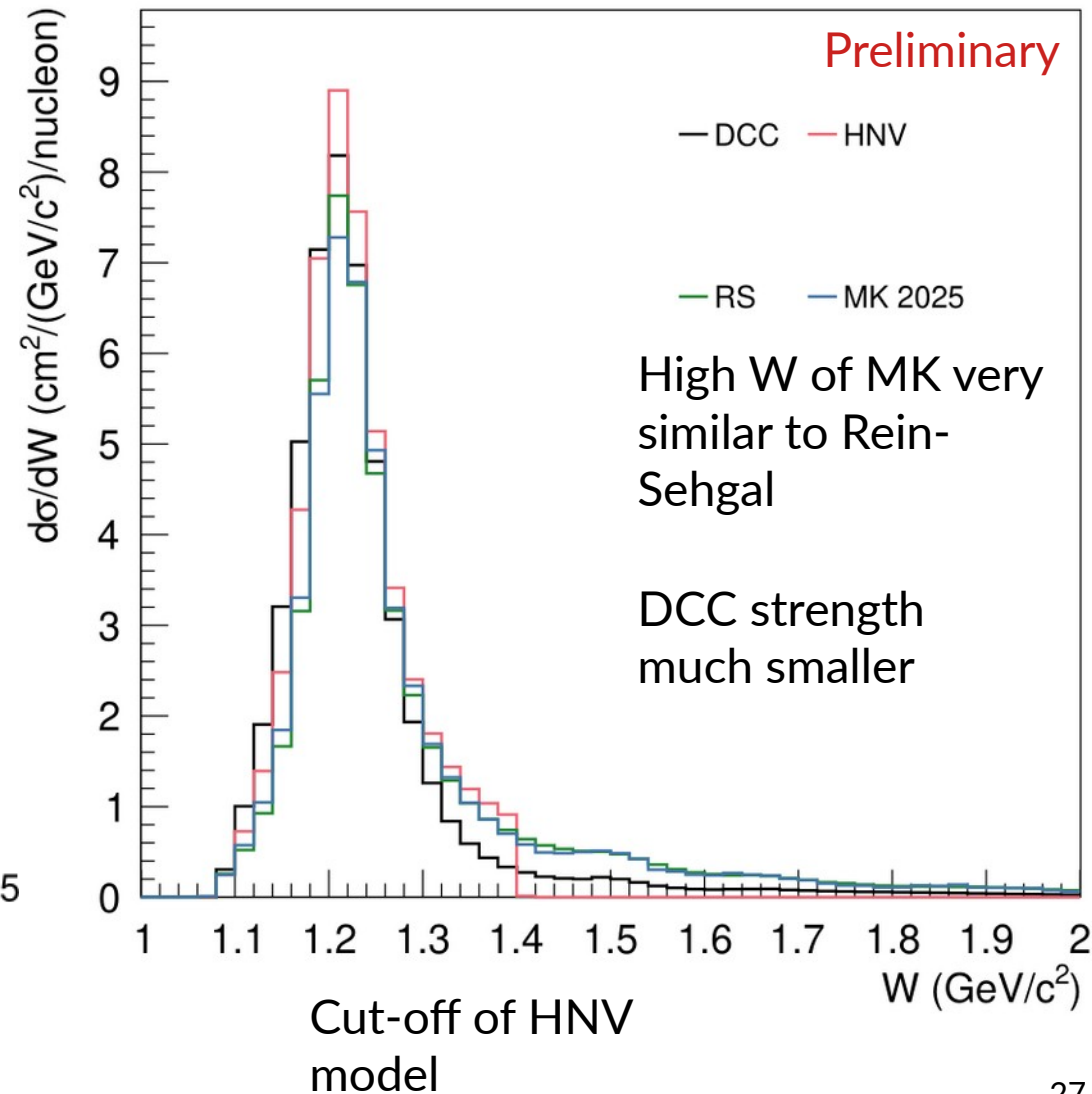
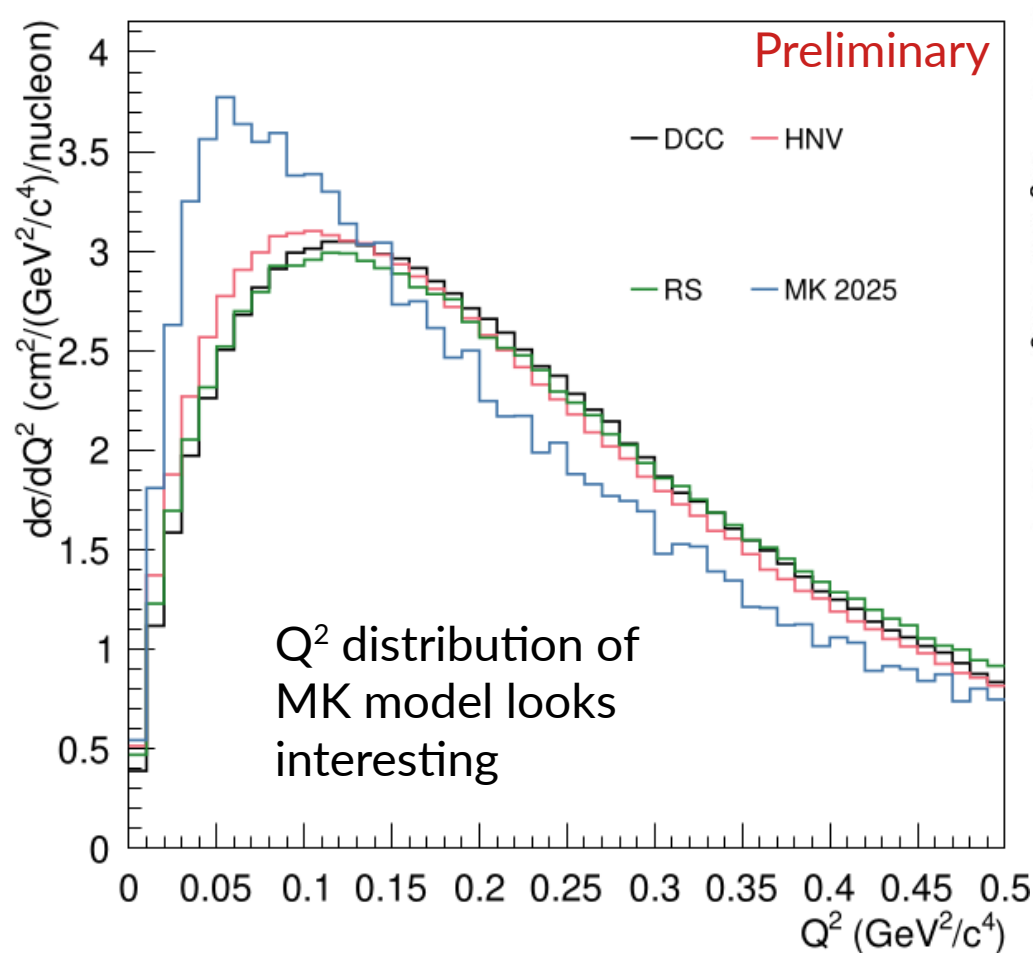
Shape comparisons (area norm)



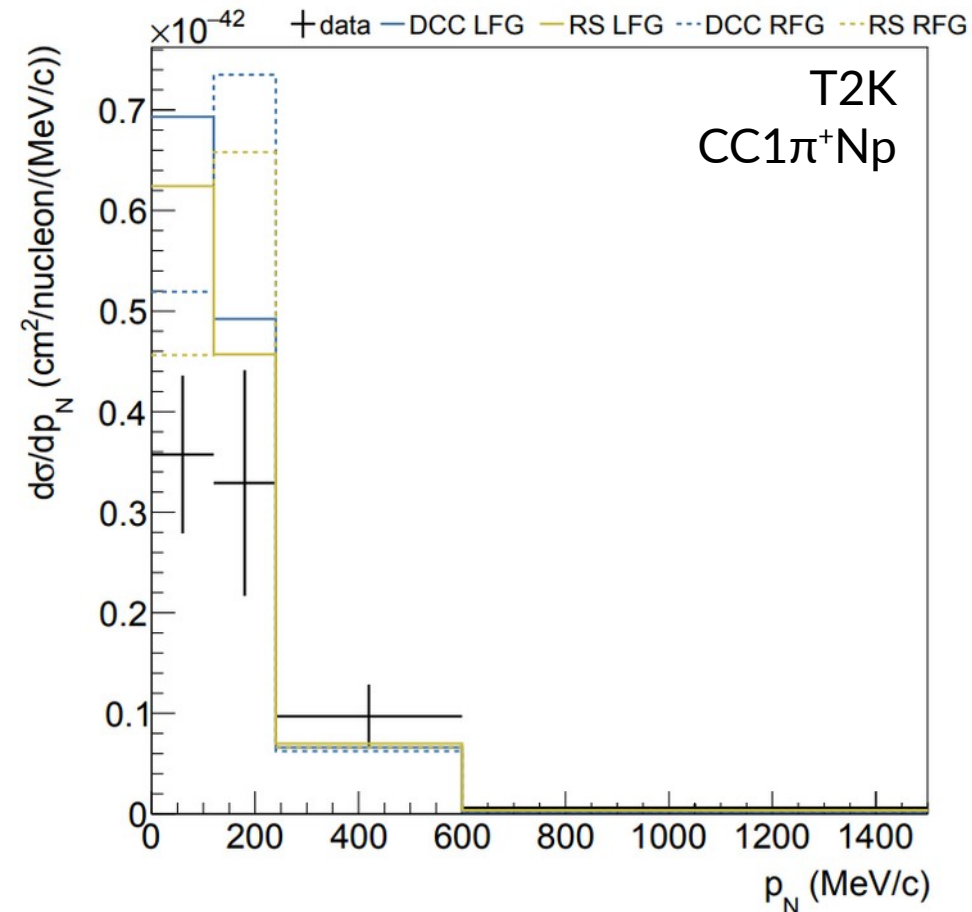
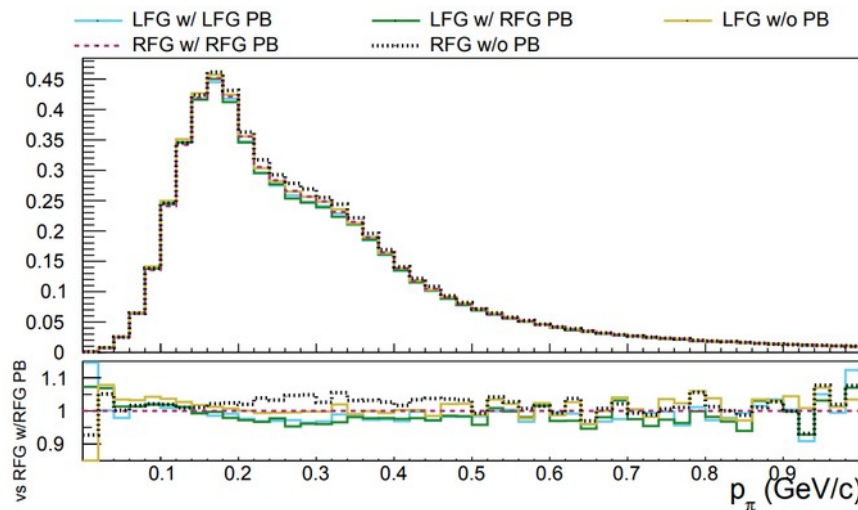
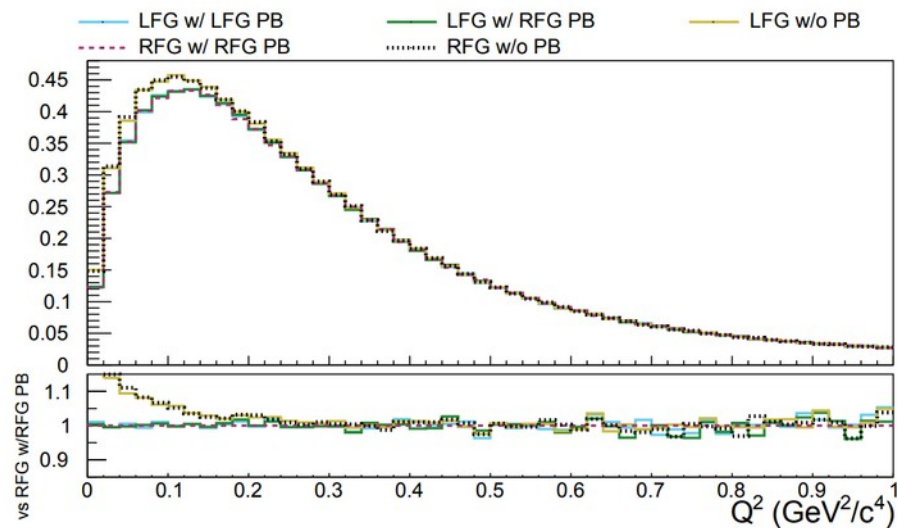
Brief comparisons

- Am almost done with validating the generators (but not confident enough to show all distributions!)

Shape comparisons (area norm)



- Also spending time refreshing the nuclear effects!
- Binding energy; LFG/RFG/SF for initial state, interaction position in nucleus and Pauli blocking (Seisho's talk)

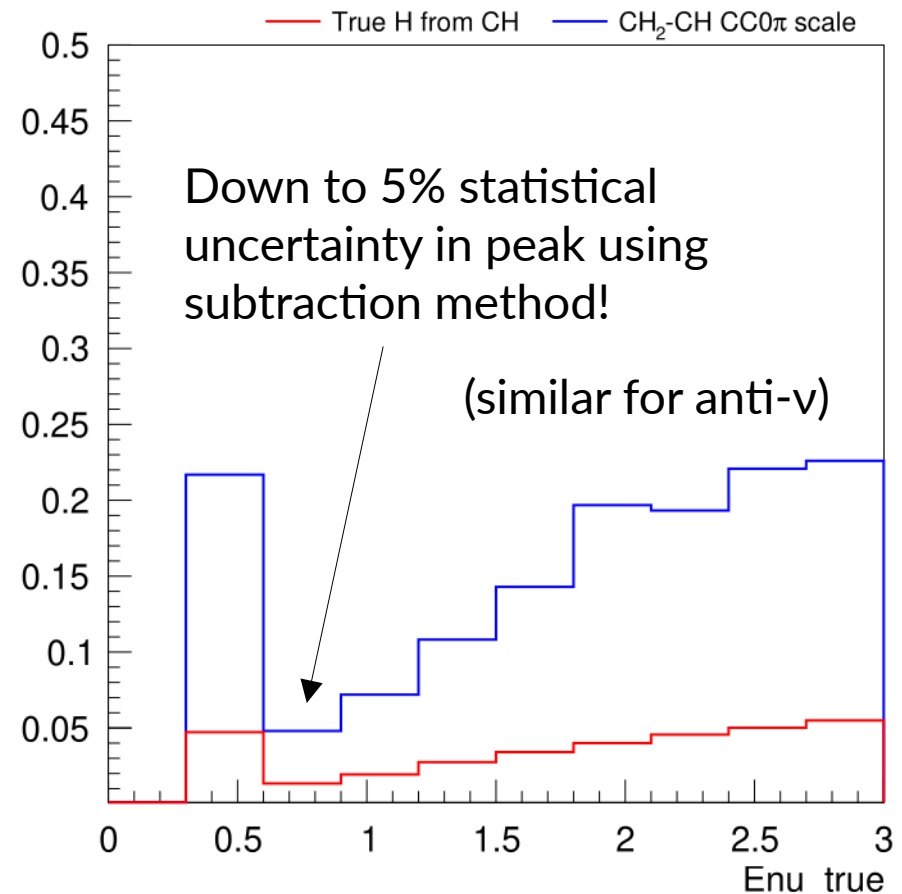
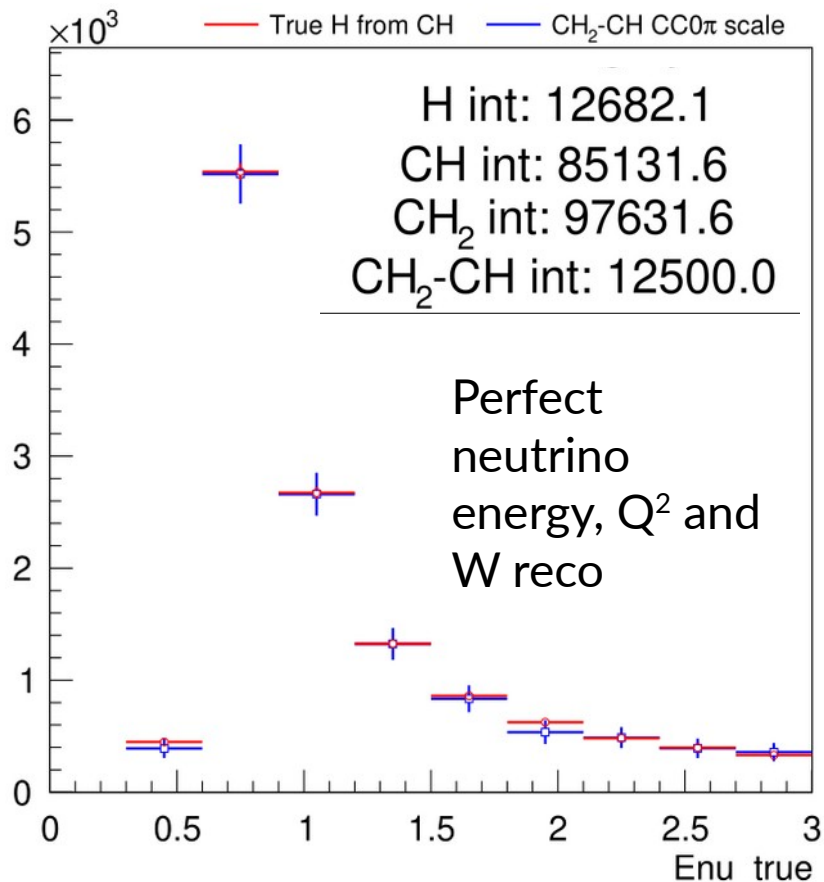


Nuclear effects?

- Lots of effects still missing
- D_2 nuclear effects, re-evaluate bubble chamber tune?
Delta in-medium effects?
 - Watch out for double counting in “cascade”
- Electron scattering comparisons
 - Available for DCC, but HNV and MK also able to do it
- Ghent model with RMF baked in
 - Mattias (see Natalie’s talk yesterday)

Shameless advertising: H detector?

- 1π modelling suffers **deeply** from lack of good quality, high statistics data. Few differential xsecs in Q^2 , W ; no 2D xsec
- No more bubble chambers, although FNAL are prototyping
- Working on H via subtraction ($\text{CH}_2\text{-CH}$) for HK-ND280 or DUNE
- Looking promising, **need to convince that community cares**



Summary

- NEUT single pion production based on Rein-Sehgal, but a fairly different Rein-Sehgal to that in GENIE
- In-house tune to make sure data is covered, not underestimating the uncertainty bad and overfitting
- Introduced new freedoms on the hadronic side of Rein-Sehgal
- Ultimately, new models had to be added
 - DCC, HNV, MK, and (soonish?) Ghent?
- Have yet to finish full validation, but desire to benchmark models soon
- Nuclear effects being studied and implemented; open to suggestions and advice
- “Hydrogen detector” for HK? Need high statistics data. **Can we do HK and DUNE without more H data?**